

Starfish

An envelope and step filter: auto-wah quack or random steps in time with the project.



Overview

Starfish is an automatic filter in the spirit of the classic 70s envelope-filter pedals and the modern sample-and-hold filter boxes. A resonant two-pole filter is swept either by your playing or by random steps in time with the project.

In ENV mode, an envelope follower moves the filter as you play - the classic auto-wah quack. In S&H mode, the filter jumps to a new random setting on every step of the beat. In BOTH, the steps are scaled by how hard you play.

The filter can be low-pass, band-pass, high-pass or notch, and the sweep can go up or down. The display shows the cutoff over time, with a line for each step.

Reach for it for

- Funk guitar and clav auto-wah
- Quacky, envelope-swept bass
- Rhythmic random filter steps on synths, pads and drum loops
- Talking, vocal-like sweeps (slow RESPONSE)
- Static resonant filtering with HOLD

Controls

Display

Cutoff display	The filter cutoff over time on a 100 Hz - 10 kHz scale, with the current value at the top right. In the step modes, a faint line marks each new step.
ENV	Lights when the envelope follower is responding to your playing.

Sweep

UP / DOWN	The sweep direction. Click the arrow at the top of the blue (UP) or red (DOWN) block; the active side lights up. UP (default) opens the filter as you play; DOWN closes it.
DEPTH	How far the sweep goes, 0 to 6 octaves (3.5 by default).
FREQ	Where the sweep starts, 60 Hz to 3 kHz (250 Hz by default).
FILTER	The filter type: LP (low-pass), BP (band-pass, the default and the classic wah sound), HP (high-pass) or N (notch).
RANGE LO / HI	HI moves the whole sweep up an octave.
PATTERN FREE / LOOP	FREE: a new random step every time. LOOP (default): a one-bar random pattern that repeats, so the steps groove with the music.
ENV / S&H / BOTH	The mode. ENV (default): your playing sweeps the filter. S&H: random steps in time with the project. BOTH: the steps, scaled by how hard you play.

Envelope

PEAK	The filter's resonance, 0.5 to 15 (4 by default). Higher settings give a sharper, vocal quack.
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SENS	How strongly your playing drives the sweep, 0 to 100% (60% by default).
RESPONSE	How fast the envelope follows your playing, 0 to 100% (40% by default): low settings are fast and snappy, high settings slow and vocal.

Steps

RATE	The step length: 1/2, 1/4, 1/8 (default), 1/16 or 1/32 of a note, following the project tempo (120 BPM if there is none).
MULT 1 / 2 / 4	Multiplies the number of steps: x1, x2 or x4.
GLIDE	Smooths the jump between steps, 0 to 100% (15% by default). 0 gives hard steps; higher settings slide from one to the next.

Output

DRIVE	Pushes the filter for a fatter, grittier sound, 0 to 100% (20% by default).
MIX	Dry / filtered balance, 0 to 100% (100% by default).
OUTPUT	Output level, -12 to +12 dB.

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB. A single click on the knob (without dragging) switches GUARD on or off; its LED sits at the knob's upper right.
GUARD	On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The LED glows brighter while it is working.

SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT (OUT)	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency, the default), or a lookahead limiter holding the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while it is clipping.

The header

HOLD	Freezes the filter where it is. The sweep stops until you switch it off.
ON	Switches the effect in or out (in by default). Out, the dry signal passes through.
PRESETS	Opens the list of factory presets; the arrows step through them and the display shows the one in use (click it for the list too). Presets set the filter controls, MIX, OUTPUT, ON and HOLD; the IN / OUT strip is left alone. The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Init	The default settings: an upward band-pass envelope wah from 250 Hz. A new instance opens on it.
Funk Up	The classic upward band-pass envelope wah.
Bass Quack (down)	Low-pass sweeping down from 900 Hz with more resonance and drive - for bass.

Ship Steps	Band-pass random steps on 1/16 notes, looping each bar, with little glide.
Random Gurgle	Low-pass random steps on doubled 1/8 notes, FREE pattern and plenty of glide, high resonance.
Envelope Steps	BOTH mode: 1/16 steps scaled by your playing.
Notch Sweep	A notch swept up by the envelope, slower response, no drive.
Slow Talker	Band-pass with high resonance and a slow response, for a talking, vocal sweep.
Clav Wah	A fast, bright band-pass wah in the HI range with extra drive.
Synth Blips	Band-pass random steps on 1/32 notes in the HI range, no glide, very resonant.
Lowpass Bloom	A wide low-pass envelope sweep from 120 Hz, gentle resonance and slow response.
Reset all to defaults	Every control back to its starting value, including the IN / OUT strip.

Tips

- Set SENS by watching the display: the cutoff line should rise well on hard notes and fall back between them.
- For a classic funk wah, use BP, UP, a PEAK around 5 and a low RESPONSE so each note quacks quickly.
- The step modes follow the project tempo and position, so S&H patterns stay in time when you start playback anywhere.
- Use LOOP for steps that repeat every bar and groove; FREE for constantly changing textures.
- Find a sweet spot by ear, then press HOLD to keep it as a fixed resonant filter.

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